CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

Hostel management has traditionally been an administrative activity centred on registers, forms and manual verification. Over the past decade, research and student projects have gradually moved from paper-based records to web-based hostel management systems, and then to sensor-based 'smart hostel' solutions that use RFID, biometrics, QR codes and Internet of Things (IoT) devices. In parallel, machine learning and natural language processing have matured in other service domains, where they are used to classify complaints, forecast demand and support decisions. However, these two lines of work have rarely been brought together in a single hostel platform.

The literature reviewed for HostelOS covers several important areas: web-based hostel administration, RFID and biometric access control, QR code-based access control and smart locks, IoT-based attendance, energy and safety monitoring, AI-based visitor detection, machine-learning classification of complaints, and a systematic review of QR based access-control architectures. In total, ten studies published between 2018 and 2026 are examined. Together, they provide the technical foundation for HostelOS and show where existing work stops short of a unified, intelligent platform that combines administration, safety, predictive analytics and resource optimization.

2.2 HOSTEL MANAGEMENT AND INTELLIGENT CAMPUS SYSTEMS

A hostel is a dense operating environment. Attendance, leave, gate movement, room allocation, fees, complaints, mess operations, inventory, maintenance, visitors and emergencies must all be handled at the same time, often by a small administrative

team. Conventional hostel management software mainly digitizes the forms that wardens and administrators already use. Its modules for fees, complaints, mess and rooms are usually built in isolation, so the data they generate is rarely used for prediction, automation or decision support.

Access control has followed a similar path. Manual registers were followed by RFID cards and biometric verification, and more recently by QR codes read through ordinary smartphone cameras. Each approach has trade-offs. RFID and biometric systems provide strong identification but need dedicated readers at every access point, which increases cost and maintenance effort. QR-based systems avoid specialized hardware, but if every routine entry and exit requires a scan, the process adds daily friction, and the security of the scheme depends heavily on how credentials are generated, validated and expired on the server.

Finally, machine learning offers techniques that suit hostel data naturally: supervised text classification for complaints, time-series and regression models for meal and inventory demand, and risk classification for maintenance. These techniques are well established in other domains, but their application to complete hostel operations has received little attention in the reviewed literature. The studies below are discussed in reverse chronological order.

2.3 REVIEW OF RELATED RESEARCH

2.3.1 QR Code-based Access Control Systems

Alsawi and Ibrahim presented a systematic literature review of QR code-based access control systems, published in the International Journal of Advanced Computer Science and Applications. The review covers studies from 2015 to 2023 drawn from major databases and organizes them into an architectural taxonomy. It also examines the security landscape of these systems and outlines future research directions. A recurring finding is that QR-based access control is attractive because it needs no specialized credential hardware, but that its security and robustness depend on design choices such as how credentials are issued and verified.

This study is relevant to HostelOS because the platform's exceptional-movement workflow is built on QR credentials. It supports the design decision that all QR validation should happen on the server, with temporary tokens that expire and are rejected and logged when reused.The limitation is that the work is survey-level only.

It identifies gaps in security and robustness but does not describe a deployed system, and it does not address how QR credentials should be tied to a student's leave or presence status in a hostel.

2.3.2 Smart Hostel Management Through the IoT

This conference paper, titled 'Smart Hostel Management Through the IoT: Reducing Energy Consumption, Securing Access, and Preventing Theft', proposes an IoT sensor network for hostels. The system uses occupancy sensing, RFID/NFC-based locks and gas leak detectors to automate energy saving and safety monitoring. It confirms that energy waste, unauthorized access and safety hazards are practical concerns in hostel environments and that automation can help address them.

The work is useful to HostelOS as evidence of the safety needs a hostel platform must serve. Its approach, however, is purely IoT- and hardware-centric. It depends on installed sensors and locks, lacks predictive analytics, and does not offer a unified administrative dashboard. HostelOS addresses safety through software, using presence tracking, SOS reporting and a digital safety map, and treats physical devices as optional.

2.3.3 Smart lock system using QR code technology

Adedoyin and Olukoya developed a smart lock system using QR code technology, published in the ABUAD Journal of Engineering Research and Development. A microcontroller-driven lock verifies a server-issued QR credential before it actuates a solenoid lock. The design shows that QR credentials issued by a server can replace physical keys and cards for door access.

The study supports the server-issued, verify-before-grant principle used in HostelOS. Its limitation is scope: it is a hardware access-control unit and is not integrated with attendance, complaints or hostel resources. A lock that only checks a credential has no knowledge of whether a student is on approved leave, so it cannot support the wider workflow that HostelOS manages.

2.3.4 Digitizes Hostel Administrative workflows

Tan's final year project at Tunku Abdul Rahman University of Management and Technology describes a full-stack web application that digitizes hostel administrative workflows and replaces manual, paper-based processes. It is a representative example of the conventional hostel management system that many institutions deploy, and it demonstrates the benefit of centralizing hostel records.

This work forms the administrative baseline against which HostelOS is positioned. The system digitizes administration only. It does not include predictive analytics or AI-based decision support, and it does not extend into safety or resource optimization. HostelOS retains the administrative core but adds an intelligence layer on top of it.

2.3.5 Smart door access control system based on QR codes

Jain and co-authors proposed a smart door access control system based on QR codes, published in the International Journal of Informatics and Communication Technology. An ESP-based door controller decodes server-issued QR codes and grants or denies physical access accordingly. The paper reinforces that low-cost microcontrollers and ordinary QR codes can provide workable access control.

Like the previous access-control studies, it concentrates on the door itself. There is no student leave or presence state behind QR issuance, so the decision to grant access is not linked to any hostel record. HostelOS differs by issuing QR credentials only after staff approval of an exceptional movement, which keeps the credential connected to an auditable request.

2.3.6 Hostel management system using artificial intelligence and IoT

Bhardwaj and co-authors proposed a hybrid smart hostel management system using artificial intelligence and IoT. Facial-recognition cameras feed an edge and cloud pipeline that performs real-time visitor detection and raises alerts. This is one of the few reviewed works that brings AI into hostel security.

The approach is camera- and edge-hardware heavy and raises privacy concerns, since it continuously processes facial data. It is also not linked to the broader administration of the hostel. HostelOS avoids dependence on facial recognition for routine operation and keeps CCTV only as an optional, controlled component.

2.3.7 IoT-based smart hostel management system

Singh and co-authors presented an IoT-based smart hostel management system at the 4th International Conference on Intelligent Computing and Control Systems (ICICCS). RFID and PIR motion sensors are networked through IoT to log attendance automatically and to control room power. The study demonstrates that attendance recording and energy control can be automated without manual registers.

Its scope is narrow, being limited to attendance and power. It does not cover complaints, mess, inventory or safety integration, and it relies on sensors deployed in the building. For HostelOS, the study confirms the value of automated presence

tracking, which the platform achieves through server-side records and exceptional-movement authorization rather than a sensor network.

2.3.8 Hostel management system with QR codes

Patel and co-authors described a hostel management system with QR codes, published in the International Journal of Computer Applications. QR codes are generated and scanned at the hostel gate for daily attendance and access verification. It is an early example of using QR codes in a hostel context.

The weakness is that QR is used for routine daily attendance, which adds friction for every student on every entry and exit. The design also lacks an exception-based or server-side authorization model. HostelOS draws a clear lesson from this: QR should be reserved for exceptional movement such as late returns, holiday outings, and medical or personal visits, while normal working-day movement needs no QR.

In contrast to the daily-attendance approach of Patel et al., HostelOS issues a QR token only after a leave or exceptional-movement request has been approved by the warden. Each token is temporary, linked to that specific request, and validated entirely on the server, so expired, invalid or reused tokens are rejected and logged. This keeps the QR mechanism secure and traceable and removes the daily scanning burden for routine movement. It also makes the scan result part of the student's leave and presence record, which the earlier QR-based hostel system did not provide.